THESIS:

Kenya should prioritise an interoperable system between M-Pesa and a CBDC to better serve its citizens and promote financial inclusion.

INTRODUCTION:

A Central Bank Digital Currency (CBDC) is virtual money backed and issued by a central bank, similar to the currencies you are already familiar with and use daily. Given the recent rise of cryptocurrencies, governments have sought to provide an official alternative. Currently, "114 countries, representing over 95 percent of global GDP, are exploring a CBDC" (Kumar), at various stages ranging from just research to a fully launched CBDC. Nigeria was one of the pioneers in the implementation of a CBDC with the launch of the e-Naira in October 2021. They hope to add 29 billion USD to their GDP over the next 10 years, with their transaction value reaching 18 million USD in just a year since launch (Kumar).

Kenya, on the other hand, is only at the research stage. Other than the "*Discussion Paper on Central Bank Digital Currency*" by the Central Bank of Kenya in February 2022, there has been little discussion on what a CBDC in Kenya would look like and how it may be executed. In this paper, I will detail the existing digital payment landscape in Kenya, pioneered by the telecommunication company Safaricom and their M-Pesa - mobile money - technology. investigate how integrating the two systems can bring advantages to consumers, firms, and the government. However, it is crucial to ensure that the integration is adequately executed and managed with care. Failure to do so could result in severe consequences for the Kenyan economy.

In this paper, I will examine the unique current digital payment structure of Kenya and discuss the potential risks and challenges that the CBK would face, as well as the many benefits of implementing a CBDC in Kenya for the CBK, consumers, firms, and the government.

1. DIGITAL PAYMENTS IN KENYA:

Kenya is unique compared to most other countries regarding the potential implementation of a CBDC because of the large existing use of the digital payment system M-Pesa. M-Pesa, meaning "mobile money," is a system developed and run by the company Safaricom since 2007. It allows digital financial transfers from one user to another. As a consumer in Kenya, you can set up an account with Safaricom for as little as 50 ksh (0.37 USD), and from then on, you can transfer money to any other Safaricom sim card with a varying transaction fee depending on the amount sent.

M-Pesa has grown rapidly and now hosts over 30 million monthly users in Kenya, which accounts for approximately 60% of Kenyans who use M-Pesa each month (Muiruri) and 96% of households (Owigar and Morawczynski). M-Pesa's success story is one of the most remarkable in the digital payments sector. It has had a significant impact on the Kenyan economy and improved the lives of millions of people living in the country. This digital payment system has played a vital role in promoting financial inclusion, boosting transaction efficiency, and streamlining business operations, ultimately reducing production costs for firms. In this paper, I will elaborate on all of these benefits of M-Pesa in greater detail.

Kenya should look to implement a CBDC to maximise the benefits of the existing M-Pesa system and how accustomed the population is to digital finance. It is uncommon for a country to have over 60% of its population using digital payments regularly, which gives Kenya a considerable head start in the race to a cashless economy. Nigeria, the only African country

with an active CBDC, aims to use its CBDC to phase out physical cash, pushing for the adoption of its eNaira (Ndeche) and increasing financial inclusion from 64% to 95% (Kumar).

2. CBDC ARCHITECTURE (Auer 88–89)

There are three main ways for the CBK to structure their CBDC: direct, indirect, and hybrid. I will first explain what each is, then the pros and cons of each.

a. Direct CBDC

The CBDC would represent a direct claim on the central bank. The central bank would then keep a record of all balances and update it with every transaction. This system therefore only has two entities at play: the central bank holding the money and the entirety that would make a legal claim on that money.

i. Pros

1. Direct CBDC allows the central bank to have complete control over the digital currency, which can help reduce the risks of fraud and money laundering.
2. This model could also enable faster and cheaper cross-border payments, which could benefit the Kenyan economy as it relies heavily on remittances from abroad.
3. Direct CBDC also allows for the implementation of smart contracts and other advanced features that could help streamline payments and reduce transaction costs.

ii. Cons

1. Implementing a direct CBDC would require significant investment in infrastructure and technology, which could be a challenge for a developing country like Kenya.
2. It could also be difficult to gain widespread adoption of the CBDC, as consumers and businesses may be hesitant to switch to a new currency. This is especially the case in Kenya, as the M-Pesa usage rates are so high and have already proven over the past 15 years to be reliable and efficient. Converting people from this system that they are familiar with and trust could be challenging, especially when they do not have full faith in the government's ability or intentions.
3. compromises in terms of the payment system's reliability, speed, and efficiency (Auer 90). Because of the
4. The implementation of a direct CBDC could also lead to the displacement of existing players in the digital payments ecosystem, which could negatively impact financial inclusion.

b. Indirect CBDC (Auer 88)

In an indirect CBDC, the consumer has a direct claim on an intermediary, with the central bank keeping track only of wholesale accounts. Consumers have no direct claim with the CB and instead deal with intermediaries, who handle all communication with retail clients and net payments. They would send payment messages to other intermediaries and wholesale payment instructions to CB. The latter settles wholesale CBDC accounts with finality.

i. Pros

1. Indirect CBDC could be a more cost-effective and efficient option for Kenya, leveraging existing digital payment providers like M-Pesa.
2. This model could also help accelerate the adoption of CBDC, as consumers and businesses may be more willing to use a digital currency that is integrated with a popular mobile money platform.
3. Indirect CBDC could also help reduce the risk of fraud and money laundering by leveraging the compliance processes and systems of existing digital payment providers.
4. relieves the CB of the responsibility for dispute resolution, KYC (know your customer), and related services as it would have no direct interaction with consumers (Auer 90).

ii. Cons

1. Implementing an indirect CBDC would require close collaboration between the central bank and digital payment providers, which could be challenging given the different objectives and incentives of these stakeholders. These potential conflicts of interest could deter the overall success and operation of the CBDC.
2. Indirect CBDC could also face resistance from traditional banks and financial institutions, which could view it as a threat to their business model.
3. CB keeps no record of individual claims; intermediaries do that, while CB keeps track of wholesale holdings. Therefore, the CB has no cash-like direct proof of any claim, so it cannot honour claims from consumers

without getting information from the intermediary. If the intermediary is under stress, determining the legitimate owner might involve a potentially lengthy and costly legal process with an uncertain outcome (Auer 90).

4. It could also be challenging to ensure interoperability between different digital payment providers, which could limit the reach of the CBDC and reduce its impact on financial inclusion.

c. Hybrid CBDC

The hybrid CBDC is a combination of the direct and indirect CBDC systems. It allows for direct claims on the central bank by consumers while allowing intermediaries to handle payments.

i. Pros

1. Hybrid CBDC could balance the benefits of direct and indirect CBDC models as it leverages existing payment infrastructure and central bank control.
2. This model could enable faster and cheaper cross-border payments, as well as advanced features like smart contracts, while also leveraging the compliance processes and systems of existing payment providers.
3. Hybrid CBDC could promote financial inclusion by enabling interoperability between different payment providers.

ii. Cons

1. Implementing a hybrid CBDC would require significant investment in technology and infrastructure, as well as close collaboration between the central bank and payment providers.

2. The CBDC's adoption and ensuring interoperability between various payment providers may both present difficulties.
3. Hybrid CBDC could also face resistance from traditional financial institutions, which could view it as a threat to their business model.

3. COEXISTENCE OPTIONS

Before exploring the potential implications of a CBDC, it's important to consider how it would coexist with M-Pesa. There are four methods of coexistence that the CBK could try: complementary services, interoperability, partnership, and competition. Some of these are similar, but there are slight variances between some that I will highlight. In the following sections, we'll explore the pros and cons of each system.

a. Complementary Services

M-Pesa and CBDC could serve complementary purposes. For example, M-Pesa could continue to serve as a mobile money platform for small-value transactions and peer-to-peer payments, while the CBDC could be used for larger transactions, cross-border payments, or as a settlement asset for financial institutions. This would allow M-Pesa to continue its role in promoting financial inclusion for the unbanked population, while the CBDC could provide a secure and efficient means of exchange for larger transactions.

In this scenario, the CBDC would be a wholesale CBDC, which means it would only be available to a select group of users, such as financial institutions. It would not be accessible to the general public.

b. Interoperability

There are three potential systems within the interoperability method: non-interoperable, partially interoperable, and fully interoperable. Interoperability refers to the ability of M-Pesa and the CBDC system to communicate and exchange information seamlessly. This allows users to transfer funds between their M-Pesa and CBDC accounts without any technical difficulties or friction.

Interoperability can be achieved to varying degrees. In a non-interoperable system, the two systems operate independently, without any communication or information exchange. In a partially interoperable system, there is some limited communication between the two systems. However, in a fully interoperable system, the two systems are fully integrated, enabling users to transfer funds between them seamlessly.

i. Non-Interoperable / Separate Systems

The first option is for the simultaneous independent existence of both CBDC and M-Pesa. The CBK would directly implement their CBDC into the economy without making any effort to integrate the CBDC with M-Pesa. M-Pesa and the CBDC would be direct competitors and work against each other to provide a lot of the same services to consumers. The CBDC would most likely be direct, as it would be the simplest to implement considering the CBDC's independence from M-Pesa and its functioning. The insistence on independence from M-Pesa could lead to the

CBK looking for other intermediary providers to deal with the consumer. This would make the CBDC compete with M-Pesa to see who can provide the same goods and services better.

ii. Partially and Fully Interoperable: Non-Direct CBDC

If the CBDC architecture is indirect or hybrid, the CBK would work with other intermediaries to provide these services. The CBDC could act as a payment option within M-Pesa. This would mean that current M-Pesa users could make payments using the CBDC in addition to their existing M-Pesa payments. The degree to which they merge and the details of their merge would determine if this is partially or fully interoperable. If the CBK sees fit to make this a fully interoperable system, then the merge would allow for seamless transfers between funds within CBDC and M-Pesa accounts without the need for separate accounts or transfer fees. Think of it as sending money from your savings account to your check-in account; since it is a transfer within the same bank, you may not be charged for the transfer. However, if your savings and check-in accounts are in different banks, even if the banks work together, there may still be fees and technical requirements to complete the transfer. The latter is an analogy for what a partially interoperable system would mean.

c. Partnership

M-Pesa and the CBDC could form a partnership where they agree that one provides a certain number of services while the other provides the rest; e.g., the CBDC provides wholesale and business-to-business transfers, while M-Pesa continues to operate in the realm of retail and day-to-day consumption. This could prove mutually beneficial for the two entities as they avoid stepping on each other's toes and prevent excessive change in the lives of everyday people, with the only change occurring in what the businesses and financial institutions use.

d. Competition

M-Pesa and the CBDC would act entirely as separate entities and compete to provide a range of financial services. If the CBDC decides to compete in retail transactions, it could see a large drop in efficiency. M-Pesa would need to continue innovating and improving its services to remain competitive.

4. Benefits of Digital Payments and Coexistence

a. Financial Inclusion:

A fully interoperable system could enhance financial inclusion by allowing more people to access digital financial services, particularly those without bank accounts, leading to increased economic activity.

b. Increased Competition in Kenya

A fully interoperable system could drive down costs and improve service quality, benefiting businesses and consumers reliant on digital payments.

c. Improved Efficiency

Interoperability could reduce the time and costs associated with transactions, benefiting businesses and consumers by enabling faster and more cost-effective payments.

d. Boost to Digital Innovation

With Kenyan FinTech already generating over $158 million in 2022 (FinTech Global), the added interoperability could spur digital innovation by encouraging the development of new services and products and creating new business opportunities for local startups and entrepreneurs.

e. Increased Government Revenue

The implementation of a CBDC could reduce the costs of printing and distributing physical currency and managing cash transactions. It could also increase government revenue by reducing tax evasion if the CBDC is designed to allow for better tracking and monitoring of transactions.

5. Constraints
    a. Time
        i. Implementation time for a fully interoperable system between M-Pesa and a CBDC in Kenya is uncertain and depends on several factors.
        ii. The technical capabilities of service providers are a key factor in determining the implementation time, including their ability to coordinate and integrate their systems.
        iii. Regulatory requirements may also impact the implementation time, with the need for new policies or updated regulations potentially extending the timeline.
        iv. Coordination between stakeholders such as the government, CBK, M-Pesa, banks, non-banks, and other financial institutions and intermediaries is essential. Their ability to collaborate and reach agreements will impact the implementation time.
        v. The availability of funding and resources will also affect the implementation timeline, with sufficient funding and resources potentially speeding up the process.
    b. The introduction of a CBDC raises fundamental issues beyond payment systems and monetary policy.

i. A general-purpose CBDC may cause instability in commercial bank deposit funding.

ii. In times of stress, people may move their funds to the central bank, which may challenge commercial banks and the central bank to manage the liquidity in the banking system.

c. 3. Introducing a CBDC could mean a wider presence of central banks in financial systems.

i. • This could lead to a greater role for central banks in allocating economic resources, which could result in overall economic losses if they are less efficient than the private sector in allocating resources.

ii. It could also move central banks into uncharted territory and lead to greater political interference.

6. Conclusion and Recommendations

In conclusion, Kenya should look to implement a fully interoperable system between M-Pesa and the CBDC. This would utilise the existing strength of M-Pesa and bring it to new heights. A fully interoperable system should be a feasible long-term goal; however, partial systems in the short run can be more viable. A hybrid CBDC is ideal as it avoids strain on CBK while utilising existing financial systems and maximising their potential. While further research into technical capacity is required, Kenya should seriously consider the possibility of a CBDC.

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